VERI-DPO: Evidence-Aware Alignment for Clinical Summarization via Claim Verification and Direct Preference Optimization

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Abstract

Brief Hospital Course (BHC) narratives must be clinically useful yet faithful to fragmented EHR evidence[1,2]. LLM-based clinical summarizers still introduce unsupported statements[3], and alignment can encourage omissions ("say-less" degeneration)[4]. We introduce VERI-DPO, which uses claim verification to mine preferences and distill them into the summarizer with DPO[5]. On MIMIC-III-Ext-VeriFact-BHC[6,7] (100 ICU patients; patient-level splits), we train a retrieval-augmented verifier to label claim–evidence pairs as Supported, Not Supported, or Not Addressed via a single-token format. The verifier scores sentence-level claims from sampled BHC candidates and aggregates margins into a coverage-aware utility to mine length-controlled, contradiction-anchored preference pairs. On held-out patients, verifier-mined preferences separate candidates by contradiction density, and VERI-DPO reduces Not Supported claim rates from 10.7% to 1.9% (local verifier judge) and from 11.6% to 6.4% (GPT-4o judge)[8], while improving validity from 76.7% to 82.5% and maintaining informative length.

Introduction

The Brief Hospital Course (BHC) is a core narrative component of the discharge summary, intended to provide a concise yet clinically actionable account of the inpatient trajectory for handoff across care settings.[1,9] As such, it must be both informative enough to support downstream decision-making and faithful to what is documented in the electronic health record (EHR).[1,9]

Automating BHC generation is difficult because the evidence needed to justify summary statements is fragmented across longitudinal notes with heterogeneous structure, variable quality, and time-dependent context.[10] A summarization model must correctly attribute diagnoses, medications, procedures, and temporal relations to the record, and avoid producing plausible but unsubstantiated statements. Even a small number of unsupported claims can have outsized impact because clinical narratives are treated as authoritative documentation and errors may propagate into follow-up care and documentation workflows.[2]

Large language models (LLMs) have advanced clinical summarization via instruction tuning[11] and retrieval-augmented generation[12], while evaluation has increasingly shifted away from surface-level similarity toward factuality-oriented assessment, motivated by evidence that overlap metrics are poorly aligned with clinically meaningful errors.[3,13] Nevertheless, reliably faithful long-form summaries remain challenging.

Two practical barriers limit progress. First, factuality supervision is expensive and granular: verifying a BHC requires decomposing it into many statements and mapping each statement to supporting evidence, which is difficult to scale with clinician annotation.[6] Second, alignment under imperfect or indirect signals can induce shortcut behavior, such as producing shorter or vaguer narratives that make fewer checkable claims (degeneration-by-omission).[4] These issues motivate methods that operationalize statement-level verification at scale while explicitly preserving informativeness.

Claim verification offers an evidence-linked framework for modeling the relationship between a claim and supporting evidence (e.g., entailment/contradiction/unknown), but in clinical text it must contend with incomplete documentation, temporal dependence, and retrieval artifacts.[14,15] Preference-based alignment methods, including Direct Preference Optimization (DPO)[5], can shape generation behavior from pairwise preferences rather than token-level labels, but require preference signals that are tightly coupled to evidence to avoid omission-style shortcuts.[4]

In this paper, we introduce VERI-DPO (*Verifier-Driven Direct Preference Optimization*), an evidence-aware alignment pipeline that uses a compact verifier as an auditable intermediate signal and distills verifier-driven preferences into the summarizer via DPO.[5] As illustrated in Figure 1, the workflow has three stages: (i) fine-tune a lightweight retrieval-augmented verifier to predict *Supported*, *Not Supported*, or *Not Addressed* for a claim paired with patient-specific evidence using a constrained single-token decision format; (ii) sample multiple candidate BHCs for an evidence-window prompt, score sentence-level claims with the verifier, and aggregate claim-level outputs into a coverage- and length-aware

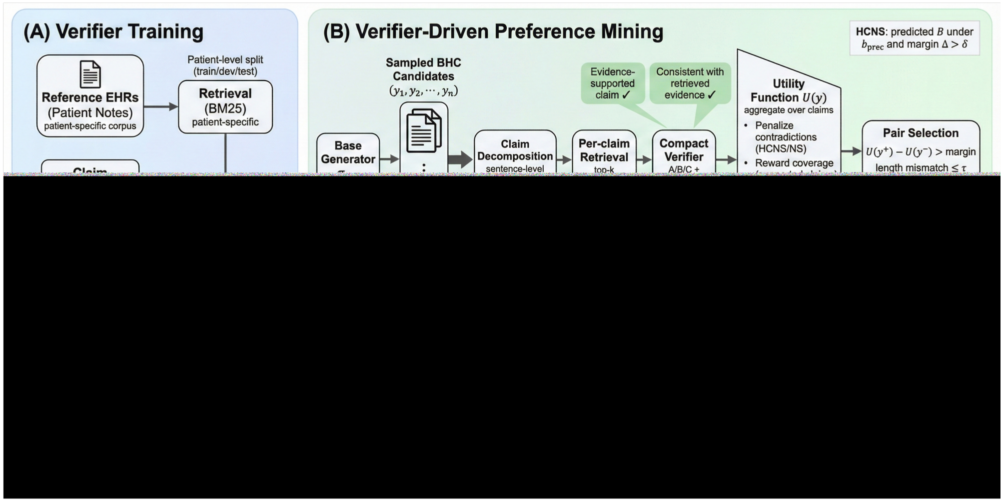

Figure 1. Overview of VERI-DPO. (A) **Verifier training:** a compact retrieval-augmented verifier predicts single-token labels `A/B/C` for each (evidence, claim) pair: `A`=Supported, `B`=Not Supported (error/hallucination), and `C`=Not Addressed (omission / not in retrieved evidence). (B) **Verifier-driven preference mining:** for each sampled BHC candidate, we decompose it into sentence-level claims, retrieve per-claim evidence, and score claims with the verifier; we aggregate claim-level signals into a utility $U(y)$ that penalizes contradictions (NS/HCNS) while discouraging degeneration-by-omission, and select (y^+, y^-) under utility and length/coverage constraints (HCNS: `B` under bias$_{\text{prec}}$ with margin $\Delta > \delta$). (C) **DPO alignment:** mined pairs are used to optimize a single-sample policy that internalizes verifier preferences, avoiding inference-time reranking.

utility that penalizes contradiction signals; and (iii) mine length-controlled, contradiction-anchored preference pairs and apply DPO to shift the summarizer toward evidence-consistent outputs, amortizing verifier guidance into a single-sample policy.

Compared with existing approaches that use factuality primarily for evaluation[3,13], our approach elevates evidence-linked verification into a scalable and inspectable training signal. The verifier produces intermediate artifacts (claim-level labels, evidence identifiers, and confidence margins) that support auditing and provide a simple calibration knob for controlling the precision–recall trade-off used in preference construction. On VeriFact-BHC (100 patients; patient-level 72/8/20 train/dev/test splits), DPO distills verifier-mined preferences into a summarizer that reduces unsupported-claim rates under both a local verifier judge and an external GPT-4o claim-level judge while maintaining validity and informative length.[8] We summarize our major contributions as follows:

- We develop a retrieval-augmented, lightweight clinical claim verifier trained with patient-level splits, and evaluate accuracy and calibration on held-out patients.
- We introduce verifier-driven preference mining for long-form clinical summaries, including high-confidence contradiction anchoring and constraints that control length and claim-coverage mismatch.
- We apply Direct Preference Optimization (DPO) on mined pairs to distill verifier supervision into a single-sample summarizer, achieving reduced unsupported-claim rates under dual-judge evaluation while mitigating degeneration-by-omission.[4]

Methods

Data and Preprocessing

Dataset. We use the MIMIC-III-Ext-VeriFact-BHC dataset[6] under credentialed PhysioNet access[16] for patient-specific factual verification of long-form clinical summaries against each patient's EHRs. The dataset is a de-identified derivative of the MIMIC-III Clinical Database v1.4.[7] It contains 100 patients (125 hospital admissions) and two BHC narratives per patient: (i) a human-written BHC extracted from the discharge summary and (ii) an LLM-generated BHC produced by summarizing the patient's longitudinal notes.[6] The patient-specific reference EHRs exclude the final discharge summary note by dataset construction and include 4,787 clinical notes in total (median 25 notes per patient; IQR 16–47; range 10–314). Note categories span 14 types, with the most frequent being Nursing (1,228), Physician (1,223), Radiology (1,134), Respiratory (471), and electrocardiogram (ECG) (306).

Proposition-level supervision. Each BHC narrative is decomposed into proposition-level units at two granularities: sentence propositions and atomic claim propositions (single predicate-like statements), yielding 13,070 proposition

Table 1. Paraphrased examples of proposition-level annotations in VeriFact-BHC. These are for illustration only and do not reproduce any individual patient record verbatim.

Source	Type	Proposition (claim)	Label	Example evidence snippet
Human	Sentence	Started broad-spectrum antibiotics for pneumonia.	Supported	Medication list notes initiation of IV antibiotics; progress note documents pneumonia treatment.
LLM	Atomic	Creatinine improved after fluids.	Not Addressed	Reference notes include fluids but no mention of creatinine trend in the provided window.
LLM	Sentence	Underwent coronary bypass surgery during admission.	Not Supported	Operative/procedure notes list no CABG; cardiology note contradicts surgery history.

instances.[6] Table 1 provides paraphrased examples illustrating the proposition format and three-way labels.

Reference EHRs and labels. Clinicians label extracted propositions as *Supported*, *Not Supported*, or *Not Addressed* with respect to each patient's reference EHRs; ground truth is established via majority vote across three clinicians with adjudication for disagreements.[6] In verifier preprocessing, we treat each labeled proposition as a candidate training instance (retaining `proposition_type` for analysis), apply claim-validity filtering before retrieval/JSONL construction, and exclude note row identifiers associated with discharge-summary sources used to derive propositions. In this dataset release, the reference EHRs already exclude the final discharge summary note; we verify that the additional row-id exclusion does not remove extra reference notes.

Research Pipeline

Objective. We aim to investigate 1) **claim verification**: given retrieved relevant evidence chunks and a claim, predict one of three labels (Supported / Not Supported / Not Addressed); and 2) **evidence-aware alignment**: generate a BHC summary that maximizes supported content while minimizing unsupported statements. Figure 1 illustrates our three-stage end-to-end evidence-aware alignment pipeline, which is described as follows:

Running example (EHR-to-BHC)

To ground the pipeline in the EHR-to-BHC setting, consider an admission where the EHR evidence window contains: (E1) a progress note documenting treatment for pneumonia with IV antibiotics; (E2) a procedure list without CABG; and (E3) labs showing creatinine values without a clear improving trend. A sampled BHC candidate may include the sentence-level claims: (C1) "Started broad-spectrum antibiotics for pneumonia" (supported by E1), (C2) "Underwent coronary bypass surgery during admission" (not supported given E2), and (C3) "Creatinine improved after fluids" (not addressed if E3 does not support the stated trend). The verifier labels each (evidence, claim) pair as `A/B/C` and these claim-level signals are aggregated into a utility that penalizes contradictions (e.g., C2) while maintaining coverage; preference mining then selects a higher-utility candidate with fewer `B` labels under length constraints, which is distilled into the summarizer via DPO.

Stage A: Retrieval-augmented verifier training

We construct training instances for verifier fine-tuning as one record per claim, pairing each proposition with retrieved, patient-specific evidence from the reference EHRs. Starting from proposition-level annotations, we merge proposition texts with clinician-derived ground-truth labels and link them to the corresponding reference notes. We split the dataset at the patient level using group-based shuffling (72/8/20 patients for train/dev/test; *dev* denotes the validation split used for model selection and calibration) and ensure that retrieval and preprocessing are performed strictly within split boundaries. After filtering, we retain 9,217/895/2,644 training instances for train/dev/test.

To prevent source overlap between labeled propositions and the retrieval corpus (i.e., avoid retrieving discharge-summary source documents used to derive the propositions), we remove such notes before indexing. Concretely, we exclude note row identifiers that match each patient's discharge-summary provenance prior to building the retrieval index. We segment each note into coarse evidence units while preserving newline structure. We then further segment these units into sentence-like spans to form retrieval candidates. We retain spans above a minimum length threshold, and also keep short but information-dense lines (e.g., labs or medication fragments). Overly long spans are truncated into fixed-size segments, and all units are lightly normalized for tokenization. For each claim, we perform two-stage lexical retrieval within the patient's reference EHR. In the first stage, we rank notes using BM25[17], a term-based relevance function that scores documents by query–term matches with length normalization, and select the top N notes (default N=35). In the second stage, we rank candidate evidence units within those notes using BM25, select the top K units (default K=50), and enforce lightweight diversity constraints by deduplicating near-identical units and capping units per originating note. We apply strict claim validity filters to remove empty strings, template fragments, and overly short/low-alphanumeric claims. Each retained instance is stored as a structured record with fields: `{proposition_id,` `subject_id, hadm_id, claim, label, evidence, evidence_meta, retrieval}`.

We fine-tune relatively small instruction-following LLMs (8B-class backbones) as claim verifiers using a next-token classification formulation with *single-token* labels. Each training instance is formatted as an instruction-following prompt containing i) a brief policy describing the three-way decision rule, ii) retrieved evidence snippets, and iii) the claim. The model is trained to output exactly one token corresponding to one of three labels: `A` (Supported), `B` (Not Supported), or `C` (Not Addressed). The loss is computed only at the label position using a restricted candidate set of the three label tokens. We train with 4-bit quantization (NormalFloat4; NF4)[18] and Low-Rank Adaptation (LoRA)[19] adapters applied to attention and multilayer perceptron (MLP) projection modules, and use gradient accumulation to achieve an effective batch size of 64 with a maximum sequence length of 2,048 tokens. Evidence is packed under a fixed token budget using up to 50 retrieved items; each item is truncated and ordered by retrieval score and time. During training, we apply *evidence dropout* by randomly removing a small fraction of retrieved evidence snippets from the prompt, which simulates missing or noisy retrieval and improves robustness to imperfect evidence retrieval. To mitigate class imbalance, we use tempered inverse-frequency class weights and label smoothing, and optionally apply a focal loss–style modulating factor that down-weights easy instances and emphasizes hard misclassified examples[20].

Given the three candidate logits for the single-token labels (A/B/C), we optionally calibrate the verifier's decision boundary by adding a scalar logit bias b to the *Not Supported* logit (class `B`) and re-decoding by argmax over the adjusted logits[21]. This simple one-parameter adjustment provides an explicit precision–recall trade-off for `B` under class imbalance and is used only for verifier calibration and downstream preference mining (it is independent of the DPO optimization itself). We sweep b over a grid on the development (validation) set and select the best bias according to the bias-tuning objective:

$$\mathcal{J}_{\text{bias}}(b) = \text{MacroF1}(b) + \lambda \cdot \text{Recall}_{\text{NS}}(b), \qquad \lambda = 0.10.$$

Here λ is a small weight that slightly prioritizes *Not Supported* recall while keeping MacroF1 as the primary selection criterion; we set $\lambda = 0.10$ based on a coarse development-set sweep over a small candidate set and found performance to be stable around this value. This provides an explicit trade-off between aggressive *Not Supported* detection and preserving performance for the majority *Supported* class. This bias is used for reporting verifier metrics; preference mining uses a separate, conservative precision-stage bias (see below) to prioritize high-precision hallucination filtering when constructing DPO pairs.

Stage B: Verifier-driven preference mining

For each prompt, we sample multiple BHC candidates, decompose each candidate into sentence-level claims, score each claim with the verifier, and aggregate claim-level signals into a summary-level utility. We extracted DPO preference pairs on the training split only, using the development split solely to select the precision-stage verifier bias ($\text{bias}_{\text{prec}}$); the test split was not accessed during the training process. For each training subject, we constructed a time-ordered corpus of evidence units from raw `NOTEEVENTS`. To prevent source overlap between proposition sources and the retrieval corpus, we excluded note `ROW_ID`s corresponding to discharge-summary sources used to derive labeled propositions before indexing the retrieval corpus. For each subject, we sampled 30 prompts; each of which included 18 evidence units (stride 8 with small jitter). We generated 8 candidates per prompt using nucleus sampling (temperature 0.8, top-p=0.95)[22]. Each candidate was cleaned, segmented into sentence-level claims, and scored for up to 24 claims. For each claim, we retrieved top-k(=50) evidence units from the subject corpus and queried the verifier under an "A/B/C" single-token formulation to obtain logits.

HCNS and two-stage biasing. To obtain high-precision contradiction anchors while maintaining sufficient mining yield, we used a conservative precision-stage bias when defining *high-confidence Not Supported (HCNS)* and a separate recall-stage bias for yield control. We define HCNS as follows:

$$\text{HCNS} \iff \hat{y}_{\text{bias}_{\text{prec}}} = \text{B} \;\wedge\; \Delta > \delta, \quad \Delta = \ell_B - \max(\ell_A, \ell_C),$$

where ℓ_A, ℓ_B, ℓ_C are the verifier (pre-softmax) logits for labels `A/B/C`. We set the margin threshold to $\delta = 0.8$ as a conservative cutoff chosen on the development split to favor precision of contradiction anchors (i.e., only treating sufficiently separated `B` predictions as HCNS); mining-level results were qualitatively stable under small perturbations of δ around this value (Table 8). For yield control, we introduced a recall-stage bias and set $\text{bias}_{\text{recall}} = +1.10$ based on a coarse development-set sweep to ensure a sufficient number of `B`/HCNS predictions for efficient pair mining. We then selected $\text{bias}_{\text{prec}}$ on the development split to trade off HCNS precision and yield under a minimum-yield constraint (at least 10 HCNS predictions), using a coarse-to-fine sweep over $[-0.8, 1.6]$. In the full run, the selected value was $\text{bias}_{\text{prec}} = -0.34$.

Candidate utility and pair constraints. Let (n_A, n_B, n_C) be the counts of verifier predictions under $\text{bias}_{\text{prec}}$ among the scored claims, and let n be the number of scored claims. We compute a summary-level utility:

$$U = \lambda_A \cdot n_A - \lambda_B \cdot n_B - \lambda_C \cdot n_C + \lambda_{\text{cov}} \cdot \min(n, n_0) - \lambda_{\text{dup}} \cdot (\text{dup\_frac} \cdot n) - \lambda_{\text{meta}} \cdot \text{meta\_hits}.$$

Table 2. Preference mining summary (training split only).

Metric	Value
Training subjects / prompts per subject / samples per prompt	72 / 30 / 8
Total mined pairs	1513
$\text{bias}_{\text{prec}}$ / $\text{bias}_{\text{recall}}$ / δ	−0.34 / +1.10 / 0.8
Frac. chosen has fewer predicted B than rejected	0.9802
Avg. #B (chosen / rejected)	0.4891 / 4.6616
Avg. score gap $U_c - U_r$	20.2958

The weights in U are treated as hyperparameters that encode an importance ranking over error types and anti-degeneration factors: *Not Supported* contradictions are penalized most heavily, *Not Addressed* receives a milder penalty (often reflecting missing evidence rather than direct contradiction), and additional terms encourage adequate claim coverage while discouraging duplication and meta text. In this paper, we empirically set λ_A=1.0, λ_B=3.0, λ_C=0.5, λ_{cov}=0.25, n_0=12, λ_{dup}=2.0, and λ_{meta}=2.0 based on development-set mining diagnostics, aiming to achieve large contradiction separation between chosen and rejected candidates while controlling coverage/length mismatch and avoiding "say-less" degeneration.

For each prompt, we select a *(chosen, rejected)* pair only if all of the following constraints are satisfied: i) the utility gap exceeds a minimum margin, $U_{\text{chosen}} - U_{\text{rejected}} \geq \tau_U$; ii) the number of scored claims differs by at most τ_n, $|n_{\text{chosen}} - n_{\text{rejected}}| \leq \tau_n$; iii) the chosen candidate contains at most τ_{HCNS} HCNS claims and at most τ_B predicted `B` claims; and iv) the rejected candidate contains at least one HCNS claim. We empirically set τ_U=2.0, τ_n=6, τ_{HCNS}=1, and τ_B=2 based on development-set mining diagnostics to obtain high-contrast preference pairs while limiting coverage mismatch. We store the mined preference pairs (x, y^+, y^-) for DPO training, where x is the prompt and (y^+, y^-) are the chosen and rejected summaries. Unless otherwise stated, we use this full preference-mining procedure (HCNS anchoring, the utility definition, and the pair constraints above) to construct the training preference dataset for DPO.

Stage C: DPO alignment

We construct a preference dataset $\mathcal{D} = \{(x_i, y_i^+, y_i^-)\}_{i=1}^N$ from the verifier-driven mining stage, where x denotes the summarizer prompt (a window of patient-specific EHR evidence units) and (y^+, y^-) are two candidate BHC summaries for the same prompt. The "chosen" summary y^+ is selected to have fewer contradiction signals (fewer verifier-predicted Not Supported claims) under length and coverage constraints, while the "rejected" summary y^- contains at least one high-confidence contradiction signal. Following the standard DPO formulation[5], let $\pi_\theta(y \mid x)$ denote the trainable summarizer policy and $\pi_{\text{ref}}(y \mid x)$ denote a fixed reference policy (the pre-DPO base model). We define:

$$s_\theta(x, y^+, y^-) = \beta\left[\left(\log \pi_\theta(y^+ \mid x) - \log \pi_\theta(y^- \mid x)\right) - \left(\log \pi_{\text{ref}}(y^+ \mid x) - \log \pi_{\text{ref}}(y^- \mid x)\right)\right],$$

and optimize the logistic loss:

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{(x,y^+,y^-)\sim\mathcal{D}}\left[\log \sigma\left(s_\theta(x, y^+, y^-)\right)\right],$$

where $\beta > 0$ controls the strength of the preference signal. We train DPO using the *Transformers Reinforcement Learning* (TRL) library's `DPOTrainer` implementation on the mined preference pairs (N = 1513; mined from the training split only). The base summarizer is `llama3.1-8b-Instruct`[23]. We apply LoRA adapters to attention and MLP projection modules (`q/k/v/o, gate/up/down`) with rank r=16, α=32, dropout 0.05. Training uses bfloat16 (BF16), gradient checkpointing, per-device batch size 1, gradient accumulation 8, max prompt length 2048, and max sequence length 3072. We sweep $\beta \in \{0.05, 0.10, 0.20\}$ and epochs $\in \{1, 2, 3\}$, and include an additional probe setting (β=0.30, epochs = 2). The learning rate is fixed at 1×10^{-5} for all runs.

To contextualize gains, we include two baselines. First, a supervised fine-tuning (SFT) baseline tests whether improvements can be explained by learning the output format rather than factual alignment: we discard the rejected response and optimize next-token log-likelihood on (x, y^+) pairs only, using the same backbone (`llama3.1-8b-Instruct`), LoRA configuration (`q/k/v/o, gate/up/down`, r=16, α=32, dropout 0.05), sequence lengths, and learning rate (1×10^{-5}). SFT uses only training-mined pairs (no test access) and does not use GPT-4o signals. Second, a verifier-guided Best-of-K reranking baseline measures how much factuality can be gained without retraining the summarizer: for each prompt, we sample K candidates from the *base* summarizer using the same decoding setup as other runs, score each candidate with the local verifier judge under the same precision-stage bias ($\text{bias}_{\text{prec}}$ = −0.34), and select the candidate with the best summary-level utility; we report K=8 as a representative compute-heavy baseline. Clinically, this corresponds to generating multiple candidate BHC drafts for the same admission (e.g., K=8 versions from the same EHR evidence window) and using an evidence-checking verifier to select the draft with fewer unsubstantiated statements while maintaining comparable coverage and length—analogous to selecting the most evidence-consistent discharge narrative among several automated drafts for clinician review.

Experimental Setup

We evaluate the verifier using macro-F1, accuracy, and class-wise precision/recall (emphasizing Not Supported). Bias calibration is selected on the development set for reporting verifier metrics; specifically, we sweep a scalar bias b added to the Not Supported logit and choose b by maximizing MacroF1 + 0.10 · $\text{Recall}_{\text{NS}}$, prioritizing Not Supported recall while maintaining overall class balance. For summarizer evaluation, we additionally report results under the same precision-stage verifier bias used in preference mining ($\text{bias}_{\text{prec}} = -0.34$) to ensure consistency with the training signal. We estimate the rate of unsupported claims using two judges: (i) the local verifier judge (same single-token ABC formulation) and (ii) an external GPT-4o claim judge operating at the claim level using only the provided evidence snippets. We additionally report validity/degeneration checks (e.g., minimum length after cleaning, duplication thresholds) and use anti-degeneration constraints in development-set selection.

For the external judge, each generated BHC is split into sentence-like claims. For each claim, we provide GPT-4o with the same evidence snippets used in the summarizer prompt (labeled excerpts `[E1]`, `[E2]`, . . .) and ask it to output a JSON label: Supported (S), Not Supported (NS), or Not Addressed (NA), along with a confidence score in [0, 1] and the evidence ids used. We define *HCNS* as an NS label with confidence ≥ 0.8 and compute:

$$\text{NS-rate}_{\text{GPT}} = \frac{n_{NS}}{n_S + n_{NS} + n_{NA}}.$$

This GPT-4o evaluation is used only for reporting and is not used to train or tune the summarizer. For model selection, we evaluate DPO checkpoints on a small development subset: we sample 8 development subjects and generate 6 prompts per subject (48 prompt instances). Each prompt consists of a time-ordered window of 18 evidence units (stride 8 with small random jitter), truncated to 220 words per unit. We generate one BHC per prompt using nucleus sampling (temperature 0.8, top-p 0.95, max new tokens 420). Each generated BHC is cleaned to remove meta text and split into sentence-like claims; after deduplication and basic validity filtering, we score up to 24 claims per BHC. For each claim, we retrieve top-k=50 evidence units from the subject-specific corpus and query the verifier using the ABC single-token scheme (A=Supported, B=Not Supported, C=Not Addressed). We apply the precision-stage verifier bias $\text{bias}_{\text{prec}} = -0.34$ and obtain per-BHC counts (n_A, n_B, n_C) over the scored claims. Let

$$n_{\text{used}} = n_A + n_B + n_C$$

denote the total number of scored claims. We report a hallucination proxy:

$$\text{NS-rate} = \frac{n_B}{n_{\text{used}}}.$$

We define HCNS by a margin threshold δ=0.8:

$$\text{HCNS} \iff \hat{y}_{\text{bias}_{\text{prec}}} = \text{B} \wedge (\ell_B - \max(\ell_A, \ell_C)) > \delta.$$

where ℓ_A, ℓ_B, ℓ_C are the verifier (pre-softmax) logits for labels `A/B/C`. We mark a generation as invalid if it is too short after cleaning, contains meta patterns, or has too few claims. We apply duplication filters based on the fraction of duplicate claims after deduplication (strict: ≤ 0.25; relaxed: ≤ 0.35). To avoid selecting degenerate models that reduce hallucinations by "saying less", we impose constraints relative to the baseline summarizer on the relaxed-valid subset: (i) minimum valid fraction ≥ 0.80, (ii) average length in characters ≥ 0.95× baseline, and (iii) average number of scored claims not worse than baseline by more than 0.5.

Results

Clinical interpretation of metrics

We decompose each generated BHC into sentence-level claims and label each claim as *Supported*, *Not Supported* (unsubstantiated or contradicted by the available EHR evidence), or *Not Addressed*. We report **NS-rate** and **#NS** as proxies for statements requiring clinician correction, and report **Valid** and output length/claim counts to ensure gains are not achieved via degeneration-by-omission ("say-less").

Verifier performance and calibration

We first report verifier performance and calibration, then present preference-mining diagnostics and DPO alignment outcomes on the development and test splits. Clinically, the verifier serves as an automated evidence-checking component that flags potentially unsubstantiated statements in a candidate BHC draft, supporting both preference mining during training and error localization during evaluation. Table 3 reports verifier performance on held-out TEST patients for two 8B backbones under our single-token (A/B/C) formulation with a DEV-selected logit bias b added to the Not Supported (class `B`) logit. We select b on the development set by maximizing MacroF1 + 0.10 · $\text{Recall}_{\text{NS}}$, which favors improved

Not Supported recall while preserving overall class balance. Under this criterion, the selected biases are $b = 0.4$ for the Llama-3.1 verifier and $b = 0.0$ for Med42 (Table 3); all TEST verifier metrics are reported using these DEV-selected biases. Llama-3.1 benefits from a positive bias that increases NS recall, whereas Med42[24] achieves higher accuracy with near-zero biasing but lower NS recall, reflecting a trade-off between contradiction sensitivity and overall class balance. For preference mining we use a separate conservative precision-stage bias ($\text{bias}_{\text{prec}} = -0.34$) to prioritize high-precision contradiction filtering, since overly large positive biases can over-trigger class B and reduce macro-F1.

Table 3. Verifier performance on held-out TEST patients. Bias b is added to the *Not Supported* logit and tuned on DEV.

Backbone	Best b (B-logit)	Dev Macro-F1	Test Macro-F1	Accuracy	NS Prec.	NS Rec.
Llama-3.1-8B-Instruct	0.4	0.5768	0.5781	0.7039	0.2214	0.4094
Llama3-Med42-8B	0.0	0.5814	0.5835	0.7496	0.2477	0.2752

Development-set model selection and baselines

To address whether improvements from preference optimization could be explained by simply learning the output format, we trained an SFT baseline (LoRA) on the screened preference pairs (training split only) using only the (prompt, chosen) targets. On a held-out development-set evaluation subset, SFT does not reduce unsupported content relative to the base summarizer: NS-rate slightly increases (0.118 vs. 0.104) and the average #Not Supported claims increases (2.28 vs. 1.92), despite comparable length and claim coverage (Table 4). In contrast, verifier-guided Best-of-K reranking (K=8) substantially reduces unsupported content without retraining the summarizer (NS-rate 0.033; n_B 0.61) and also improves validity (0.917). The selected DPO checkpoint achieves the lowest NS-rate and n_B among single-sample methods, while improving validity and increasing output length and the number of scored claims relative to the baseline, suggesting that the reduction in unsupported content is not explained by shorter or less informative generations.

We conducted a DPO hyperparameter sweep using the mined preference pairs (1513 pairs) and selected checkpoints using the development-set evaluation subset protocol. Table 5 reports the top sweep settings ranked by NS-rate; across the grid, moderate β values (0.05–0.10) with 2–3 epochs perform best under the anti-degeneration constraints, while larger β tends to reduce validity or increase instability.

Test-set main results under dual judges

We next evaluate all trained and baseline systems on the held-out TEST split (120 prompts; relaxed validity) under two judges: our local verifier judge and an external GPT-4o claim-level judge. Table 6 summarizes the main TEST results. Under the local verifier judge (Med42-8B[24]; ABC single-token; $\text{bias}_{\text{prec}} = -0.34$), DPO reduces hallucinations substantially: NS-rate drops from 10.7% to 1.9% and #NS from 1.98 to 0.36. Under GPT-4o, the absolute rates are higher due to stricter labeling and more NA assignments, but the trend is consistent: NS-rate decreases from 11.6% to 6.4% and #NS from 2.14 to 1.26. Importantly, validity improves (76.7% → 82.5%) and supported-claim counts increase, indicating that gains are not driven by "say-less" degeneration. Finally, SFT does not improve factuality over the base model, whereas verifier-guided Best-of-K reranking (K=8) reduces NS but requires multiple samples at inference time; DPO achieves the best factuality as a single-sample policy.

Clinical interpretation of main results

In practical terms, the base summarizer produces 1.98 Not Supported claims per BHC under the local verifier judge, while our DPO-aligned model reduces this to 0.36, with increased supported-claim counts and longer outputs. Under the external GPT-4o judge, the same trend holds (2.14 → 1.26 Not Supported claims per BHC), suggesting the improvement is not an artifact of a single judge. For example, a Not Supported claim may state that a patient "underwent coronary bypass surgery during admission" when procedure notes do not document CABG, or assert a lab trend (e.g., "creatinine improved after fluids") when the available evidence window does not support the stated change.

Preference mining diagnostics and robustness

To validate that the verifier-driven preference signal is not an artifact of candidate sampling or brittle hyperparameter choices, we report mining-level diagnostics computed on the **training split only** mining run (no test access). We analyze how different pair-selection strategies affect contradiction separation and length/coverage mismatch, and evaluate stability under small perturbations of δ and $\text{bias}_{\text{prec}}$ (Tables 7–8). Random pairing produces minimal separation in predicted contradictions between chosen and rejected ($\Delta\#B \approx 0.10$), indicating a weak supervision signal. Removing HCNS anchoring increases contradictions in chosen outputs (higher $\#B_c$), consistent with greater false-positive contamination. Removing length/coverage incentives maintains contradiction separation but shifts chosen outputs toward shorter generations (negative Δchars and Δn_{used}), consistent with "say-less" degeneration. Under the full strategy, length/coverage mismatch between y^+ and y^- is small ($\Delta n_{\text{used}} = +0.290$ and Δchars=+22.9), indicating that the preference signal is not driven by "saying less" but by reduced contradictions. In the ablations below, we modify one

Table 4. Development-set evaluation subset (48 prompts; relaxed validity). Lower NS-rate and n_B are better; higher validity, n_A, and length indicate reduced "say-less" degeneration. **Bold** denotes the best value in each column and <u>underline</u> denotes the second best.

Run	Valid	NS-rate	n_B	n_A	n_C	Frac. n_B>0	Claims	Chars
Base (single)	0.7708	0.1041	1.9189	14.1892	1.4054	0.6486	17.5135	1805.3
SFT (single)	0.6667	0.1181	2.2813	<u>15.1563</u>	0.8125	0.5938	18.2500	<u>1903.0</u>
Verifier rerank (Best-of-8)	**0.9167**	<u>0.0334</u>	<u>0.6136</u>	17.7727	<u>0.7273</u>	<u>0.3636</u>	**19.1136**	1876.2
DPO (Ours, best)	<u>0.8958</u>	**0.0163**	**0.2791**	**18.3256**	**0.2558**	**0.2093**	<u>18.8605</u>	**2071.9**

Table 5. Top-10 DPO sweep settings on Development-set (relaxed validity), ranked by lowest NS-rate, lowest n_B, and highest n_A.

Run	β	Ep	Valid	NS-rate	n_B	n_A	Chars
`beta0p05_ep3`	0.05	3	0.8958	0.0163	0.2791	18.3256	2071.9
`beta0p1_ep3`	0.10	3	0.8542	0.0267	0.4878	18.7073	2105.4
`beta0p05_ep2`	0.05	2	0.8750	0.0281	0.5000	18.0000	2022.1
`beta0p1_ep2`	0.10	2	0.8333	0.0313	0.5750	17.6000	2019.7
`beta0p2_ep3`	0.20	3	0.8333	0.0323	0.6500	17.7250	2035.8
`beta0p3_ep2`	0.30	2	0.7917	0.0411	0.6842	16.9737	1994.9
`beta0p2_ep2`	0.20	2	0.8750	0.0570	0.8571	17.1429	1927.5
`beta0p2_ep1`	0.20	1	0.8542	0.0700	1.2195	15.5366	1850.1
`beta0p1_ep1`	0.10	1	0.7708	0.0724	1.2703	16.0541	1912.1
`beta0p05_ep1`	0.05	1	0.7500	0.0831	1.3889	15.6944	1792.4

component at a time while keeping all other mining settings fixed.

Because δ (HCNS margin threshold) and $\text{bias}_{\text{prec}}$ directly affect which candidates qualify as "high-confidence" contradictions, we conduct a mining-level robustness check (no DPO retraining). For each prompt, we reuse the same sampling protocol and apply identical validity filters and pair constraints, while varying *only* δ or $\text{bias}_{\text{prec}}$. Table 8 shows that pair yield and contradiction separation remain stable across a small grid: the mined dataset consistently exhibits a large gap in predicted contradictions between rejected and chosen summaries ($\Delta\#B \approx 3.9$–4.4), with similar pair counts (1.50k–1.54k), suggesting the preference signal is not a fragile artifact of a single threshold choice.

Limitations

There are several limitations to this investigation that should be acknowledged. First, our experiments were conducted on a small cohort (100 patients) from a single intensive care unit (ICU)-focused dataset. Documentation practices, note availability, and clinical language vary across institutions and specialties, and both verifier behavior and alignment effects may shift under domain change. Second, the approach is retrieval-dependent: both verification and preference construction assume that relevant supporting evidence can be retrieved from the reference corpus. Missing, poorly indexed, or temporally misaligned evidence may lead to incorrect *Not Addressed* assignments or missed contradictions, and these retrieval artifacts can propagate into mined preferences and downstream DPO updates. Third, evaluation relies on automated judges (a local verifier and a GPT-4o claim-level judge) rather than clinician end-to-end review of generated BHCs; while the dual-judge setup provides complementary perspectives, absolute NS-rates should be treated as proxies, and judge disagreement may reflect differences in calibration, evidence sensitivity, or handling of clinical uncertainty. Finally, several design choices in the evidence-window construction and retrieval/verification pipeline (e.g., evidence-window size and stride, evidence packing budget, retrieval depth, and the number of scored claims) were selected empirically. A rigorous sensitivity analysis over these hyperparameters would require a large-scale, computationally expensive grid search. However, we expect moderate variations of these hyperparameters to be unlikely to overturn the main trends, as suggested by the robustness checks in Table 8.

Conclusion

We introduced VERI-DPO, a verifier-driven DPO framework for evidence-aware alignment in long-form clinical summarization. By training a compact retrieval-augmented claim verifier and using its claim-level predictions and confidence margins to curate preference pairs under explicit coverage and length constraints, VERI-DPO provides scalable, inspectable supervision for DPO. Empirically, verifier-mined preferences separate candidates by contradiction density, and DPO distills this signal into a single-sample summarizer that reduces unsupported claims under both a local verifier and an external GPT-4o judge while preserving validity and informative length, mitigating the common "say-less" failure mode or omission. Beyond factuality gains, VERI-DPO produces evidence-linked intermediate artifacts—claim labels, confidence margins, and retrieved evidence identifiers—that enable efficient error localization, clinician review,

Table 6. Main Results on the test split (120 prompts). Comparison of baselines vs. DPO across two judges. **NS-rate**: Fraction of claims labeled Not Supported (lower is better). **#NS**: Average count of Not Supported claims per summary. **Valid**: Percentage of summaries meeting length and format constraints. **Bold** denotes the best value in each column and underlining denotes the second best. *Key Finding:* SFT fails to reduce hallucinations compared to the Base model, whereas DPO achieves the lowest NS-rate under both judges while maintaining high validity and output length.

	Local Verifier Judge (L)			**GPT-4o Judge (G)**			**Overall**	
Method	**NS-rate ↓**	**#NS ↓**	**#Supp. ↑**	**NS-rate ↓**	**#NS ↓**	**#Supp. ↑**	**Valid ↑**	**Chars ↑**
Base (Llama-3.1)	10.7%	1.98	15.0	11.6%	2.14	12.0	76.7%	1855
SFT Baseline	10.1%	1.92	14.4	10.0%	2.04	12.8	64.2%	1865
Verifier Rerank (K=8)	3.4%	0.69	18.9	8.3%	1.76	14.4	**85.0%**	1900
DPO (Ours)	**1.9%**	**0.36**	**19.1**	**6.4%**	**1.26**	**14.6**	82.5%	**2159**

Table 7. Ablation diagnostics for preference mining on the training split (all strategies evaluated under the same verifier with $\text{bias}_{\text{prec}} = -0.34$). $\#B_c$ and $\#B_r$ denote the average number of verifier-predicted B (Not Supported) claims in the chosen and rejected summaries, respectively. We define $\Delta\#B = \#B_r - \#B_c$ (higher indicates better contradiction separation). Δchars and Δn_{used} are computed as (chosen − rejected), where n_{used} is the number of scored claims; values closer to 0 indicate smaller length/coverage mismatch. Positive values mean the chosen summary is slightly longer / uses more scored claims than the rejected summary.

Strategy	**#Pairs**	**$\#B_c$ ↓**	**$\#B_r$**	**$\Delta\#B$ ↑**	**Δchars** (closer to 0 is better)	**Δn_{used}** (closer to 0 is better)
Full (default)	1533	0.410	4.295	3.886	+22.9	+0.290
Random (no-verifier)	1899	2.364	2.466	0.102	+5.4	-0.177
No-HCNS anchoring	1882	0.947	4.921	3.973	+32.5	+0.451
No length/coverage incentives	1576	0.301	4.644	4.343	-22.1	-1.637

and post-hoc auditing without inference-time reranking.

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Table 8. Preference-signal stability for training-split preference mining under small perturbations of δ and $\text{bias}_{\text{prec}}$. $\#B_c$ and $\#B_r$ denote the average number of verifier-predicted B (Not Supported) claims in the chosen and rejected summaries, respectively. $\Delta\#B = \#B_r - \#B_c$ (higher indicates better contradiction separation). Δchars is computed as (chosen − rejected); values closer to 0 indicate smaller length mismatch (positive values mean the chosen summary is slightly longer). *Absolute #pairs may differ from Table 2 due to re-sampling under a different random seed; trends are reported for robustness.*

Setting	**#Pairs**	$\Delta\#B \uparrow$	$\#B_c \downarrow$	$\#B_r$	**Δchars** (closer to 0 is better)
Vary δ (*fixed* $\text{bias}_{\text{prec}} = -0.34$)					
$\delta = 0.6$	1503	4.140	0.456	4.596	+19.8
$\delta = 0.8$ (base)	1520	4.212	0.508	4.720	+14.3
$\delta = 1.0$	1524	4.258	0.543	4.801	+12.4
Vary $\text{bias}_{\text{prec}}$ (*fixed* $\delta = 0.8$)					
$\text{bias}_{\text{prec}} = -0.54$	1536	3.930	0.438	4.368	+12.5
$\text{bias}_{\text{prec}} = -0.34$ (base)	1520	4.212	0.508	4.720	+14.3
$\text{bias}_{\text{prec}} = -0.14$	1495	4.434	0.555	4.989	+19.6

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